

Results

01 • PRINCIPAL COMPONENT ANALYSIS (PCA)

reduce variables to components

Table 1. Variance explained

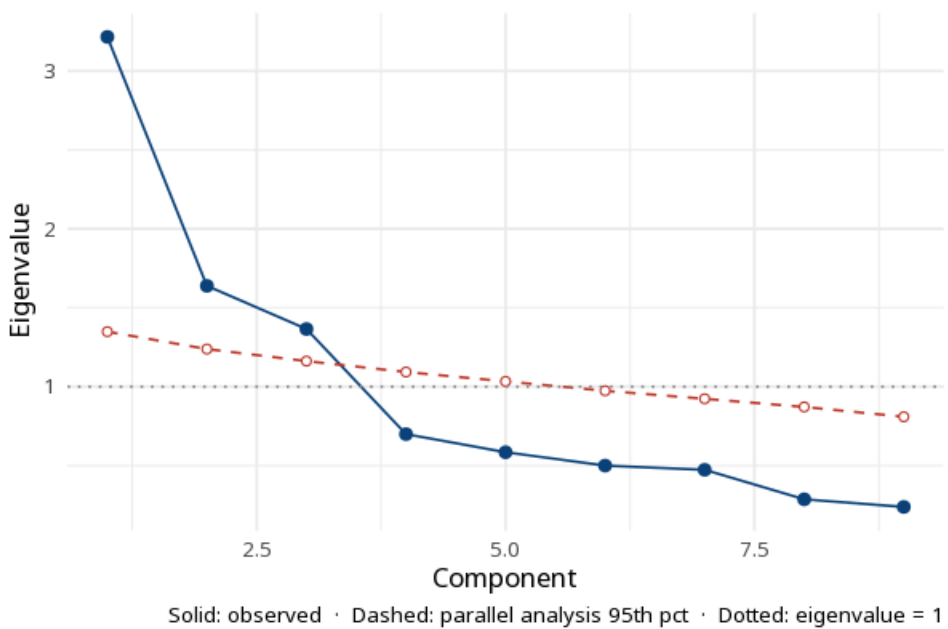
Component	Eigenvalue	% variance	Cumulative %
PC1	3.216	35.7	35.7
PC2	1.639	18.2	53.9
PC3	1.365	15.2	69.1

Table 2. Component loadings

Variable	PC1	PC2	PC3
x1	.66		−.37
x2	.39		−.62
x3	.48	.33	−.54
x4	.77	−.45	
x5	.74	−.48	
x6	.77	−.43	
x7	.35	.50	.59
x8	.45	.61	.33
x9	.59	.51	

Component loading columns expand to the number of retained components; loadings are correlation-scaled (eigenvector × $\sqrt{\text{eigenvalue}}$); loadings $|< .32|$ suppressed. PCA is data reduction — components are weighted composites, not latent factors; communalities are not reported (Jolliffe & Cadima, 2016; Frick et al., 2025).

Figure. Component retention



How to read this test

PCA compresses correlated variables into a few components. The variance explained (Table 1) shows how much information each component retains; the component loadings (Table 2) show how strongly each variable contributes. Loadings are correlation-scaled ($\text{eigenvector} \times \sqrt{\text{eigenvalue}}$) so they share the unit of a correlation coefficient. PCA is data reduction — components are weighted composites of the observed variables, not reflective latent factors; do not interpret them as constructs and do not report communalities (which belong to factor analysis, not PCA). For a latent-construct model use EFA (psych::fa) or CFA (lavaan). The scree plot overlays the parallel-analysis 95th-percentile threshold — retain components where the observed eigenvalue exceeds that line; the Kaiser eigenvalue > 1 rule is shown for reference but tends to over-extract (Zwick & Velicer, 1986). Cumulative variance $\geq 70\%$ is a common but explicitly subjective target (Jolliffe & Cadima, 2016).

APA template: PCA (correlation matrix; parallel analysis) retained 3 components explaining 69.1% of total variance.

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